



Institute of Information and Communication Technologies, Electronics and Applied Mathematics
@UCLouvain (Belgium)

invite applications for

A DOCTORAL and a POSTDOCTORAL FELLOWSHIP in Data Mining and Visualization Techniques applied to MIGRATION STUDIES
starting in September 2018 and ending in August 2022

Job Description. - INGI (the Computer Science Department part of the ICTEAM institute of the UCLouvain) obtained financing to conduct research on the application of data mining and visualization techniques in **“Migration Studies”**. In September 2018, a new **project** on “New approaches to understanding and modeling global migration trends” (GLOBMIG) will start. The objectives of this project are to gain an understanding in the long-run root drivers of international migration, to produce predictions of migration, population, and to use knowledge bases to assess the effectiveness and policy coherence of legal frameworks. Research should go beyond the state of the art in combining traditional and new sources of data (e.g., Big Data on cell-phone owners’ mobility, worldwide opinion surveys on migration intentions, geo-referenced data on population changes, comparative data on immigration policies). Core topics in this project will be the development of new methodologies, algorithms and tools for processing and analyzing this wide range of data. The candidates will be expected to contribute to this project by working in close collaboration with **Prof. Pierre Schaus** (Computer Science), **Siegfried Nijssen** (Computer Science) and **Prof. Frédéric Docquier** (Economics). The candidates should have conducted promising theoretical or empirical research in either data mining, machine learning, data-visualization or combinatorial optimization.

General conditions. - Post-doc candidates should normally have successfully defended their PhD thesis in Data Mining/Machine Learning/Data Viz or Mathematical Optimization before starting the fellowship. They must have a proven record of high quality research in data science. Doctoral and Post doctoral candidates should not have stayed more than 12 months in Belgium during the last three years. The starting date of the fellowship mentioned above can be somewhat postponed. Appointments are research positions for two years, with a possibility to extend for two more years. Net salary is of the order of 2400 euros per month for the Post-Doc and 1800 euros per month for the Doc. Members of the EU are covered by the Belgian social insurance scheme. Non-members are only insured for health care and can receive child benefit. The candidate will be located in the INGI department <https://uclouvain.be/en/research-institutes/icteam/ingi>. Post-doc Candidates who will defend their PhD thesis in the course of 2018 can be considered as well. The net salary will be smaller as long as the PhD is not completed.

Applications. - Should be sent to pierre.schaus@uclouvain.be and siegfried.nijssen@uclouvain.be not later than **June 30, 2018**. The core of this email should contain the following information: (i) your name, (ii) your current affiliation (university, research center, etc.), (iii) your main fields of research in a number of keywords, (iv) two names and addresses (including e-mail) of persons who can be contacted for reference on the candidate, (v) an URL-link at which a sample of research papers in pdf format can be downloaded for the post-doc position / Master thesis for the PhD position. The two following PDF-files should be sent in attachment: (i) a complete curriculum vitae, and (ii) a statement of research interests.